

CENTRAL POWER RESEARCH INSTITUTE, BENGALURU

Short Circuit Laboratory - TEST REPORT

Test report / series no.	CPRIBLRSL25T1654
Sample code no.	HVD25S0847
Customer	A.P. Transformers, SP24 & 95H/1 (SP), Ambattur Indl Estate, Chennai - 600058
Date(s) of test	28-10-2025 to 18-11-2025
Reference standard	IS 1180 (latest: yes)
Tests performed	Type test (Dynamic ability to withstand short circuit (with thermal short); Temperature rise test; Pressure test (IS 2026); Vacuum test (IS 1180); No load current at 112.5%; Oil leakage test), plus routine tests
Witness	Mr. Vivekanand Krishna (customer representative)
Report prepared by	P. Naveenkumar (Test Engineer)
Decision rule	Yes - (i) compliance per standard; MU reported if required
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1. Description of test sample

Sample	3 Phase Distribution Transformer
Rating	250 kVA / 11 kV / 433 V, 3PH, 50 Hz, copper wound, non-sealed, outdoor, tapping +5% to -10% (7 taps)
Serial no.	1098
Drawing nos.	APT/250-11/001, APT/250-11/001B
Voltage / phases / freq	11000 V / 433 V, 3 ph, 50 Hz, Dyn11, ONAN
Taps	+5% to -10% in steps of 2.5% (7 taps)
Insulation levels	Um 12 kV; HV LI 75 kV / AC 28 kV rms; LV AC 3 kV rms
Impedance (75 C)	4.5 %
Guaranteed losses	980 W at 50% load; 2930 W at 100% load
Oil volume / manufactured	460 l / 10/2025
Construction	Oil immersed, non-sealed, outdoor; LV helical/spiral copper, HV crossover copper; circular coils; CRGO core; tank plain/corrugated

2. Summary of results

Test	Result obtained	Requirement	Verdict
Short-circuit withstand (dynamic + thermal)	HT: 7.53 kA rms / 16.8 kA pk; LT: 7.51 kA rms / 16.36 kA pk; NT: 7.63 kA rms / 16.79 kA pk	Within +/-10% of required; no abnormality	PASS
Temperature rise	Top oil 26.1 K; HV wdg 39.7 K; LV wdg 34.3 K	Oil 35 K; winding 40 K	REVIEW*
Total loss (75 C)	2623.21 W (100%); 908.09 W (50%)	2930 W; 980 W	PASS
Impedance	see section 3.2	4.5 % +/-10%	PASS
No-load current at 100% / 112.5%	1.689 A / 4.261 A	<=2% / <=5% of rated	PASS
Voltage ratio (7 taps)	max dev. 0.39%	+/-0.5%	PASS
Dielectric routine tests	No disruptive discharge, withstood	no disruptive discharge	PASS
Pressure / vacuum / oil leakage	Leakage: No leakage at any point; deflection 1.08 mm (pressure), 2.87 mm (vacuum)	No leakage; deflection as logged	PASS

3. Detailed results

3.1 Winding resistance (HV ohm / LV milli-ohm), before / after short circuit

Tap/winding	Before: R1, R2, R3	After: R1, R2, R3
HV tap H	4.2028, 4.1822, 4.1874	4.22, 4.1988, 4.2092
HV tap L	3.5476, 3.5398, 3.5458	3.5612, 3.554, 3.5622
HV tap N	3.9736, 3.9558, 3.969	3.989, 3.9734, 3.9864

Tap/winding	Before: R1, R2, R3	After: R1, R2, R3
LV	3.8716, 3.865, 3.9074	3.897, 3.8844, 3.9248

3.2 Losses and impedance (reference temperature 75 C)

Tap	%Z	%X	%X chg	Load loss W	Stray W	Total 100% W	Total 50% W	Isc rms/pk kA
NTBT	4.38	4.299	-	2102.74	107.62	2485.92	908.09	7.61/16.73
NTAT	4.387	4.306	-0.163	2101.21	106.21	2484.47	906.87	-
HTBT	4.442	4.365	-	2056.03	111.77	2439.21	-	7.5/16.61
HTAT	4.446	4.369	-0.092	2054.67	110.09	2437.93	-	-
LTBT	4.487	4.397	-	2240.03	118.6	2623.21	-	7.43/16.2
LTAT	4.496	4.406	-0.205	2239.94	118.58	2623.2	-	-

No-load loss: 383.18 W (before), 383.26 W (after).

3.3 No-load current / loss

Condition	Avg V	I1, I2, I3 (A)	Avg I	Watts	f Hz
BT	250.03	1.907, 1.445, 1.714	1.689	382.92	49.95
AT	250.01	1.893, 1.438, 1.73	1.687	383.3	49.99
112.5%	281.356	4.681, 3.937, 4.165	4.261	553.73	49.94

3.4 Routine tests (before / after short circuit)

Test	Before	After
IR HV-Earth (Gohm)	4.19	6.11
IR HV-LV (Gohm)	6.56	7.81
IR LV-Earth (Gohm)	1.22	1.35
Induced over-voltage	866 V, 100 Hz, 60 s, 1.58 A	1.66 A - No disruptive discharge, withstood
HV / LV power-frequency	28 kV / 3 kV for 60 s	No disruptive discharge, withstood
Ambient / RH	26.8 C / 79.8%	26.7 C / 71.0%
Voltage ratio tap 1	46.38, 46.36, 46.26	46.37, 46.37, 46.26
Voltage ratio tap 2	45.22, 45.22, 45.12	45.23, 45.23, 45.13
Voltage ratio tap 3	44.1, 44.09, 44.06	44.11, 44.09, 44.07
Voltage ratio tap 4	43.02, 43.01, 42.96	43.03, 43.05, 43.0
Voltage ratio tap 5	41.92, 41.92, 41.92	41.93, 41.92, 41.92
Voltage ratio tap 6	40.82, 40.86, 40.86	40.81, 40.86, 40.81
Voltage ratio tap 7	39.73, 39.72, 39.72	39.73, 39.72, 39.72

3.5 Short-circuit test (31-10-2025; As after routine test)

Osc	Tap	Peak kA	RMS U	RMS V	RMS W	Avg	Dur s	Note
S001	NT	5.32	2.07	2.11	2.07	2.08	0.54	calibration
S002	NT	16.8	7.45	7.59	7.53	7.53	0.54	
S003	NT	16.82	7.57	7.76	7.65	7.66	0.54	
S004	NT	16.74	7.61	7.85	7.68	7.71	0.54	
S005	HT	16.89	7.48	7.45	7.62	7.52	0.54	
S006	HT	16.78	7.53	7.46	7.64	7.54	0.54	
S007	HT	16.73	7.51	7.44	7.65	7.53	0.54	
S008	LT	16.49	7.68	7.56	7.64	7.63	0.54	
S009	LT	16.34	7.47	7.4	7.44	7.44	0.54	
S010	LT	16.26	7.5	7.39	7.47	7.45	0.54	
S011	NT	-	7.53	7.58	7.55	7.55	2.01	thermal

During / after test: No abnormalities / No abnormalities. Untanking: Conductor, core & clampings: no visible damage; spacers/support blocks: intact; oil: clear.

3.6 Temperature rise (09-11-2025 to 10-11-2025; short-circuit method, LT tap, 14.58 A, injected 2623.2 W)

Hour	Top oil C	Bottom oil C	Mean ambient C	Oil rise K
0	23.1	21.65	21.13	1.97
1	29.8	23.25	20.87	8.93
2	34.55	26.0	20.72	13.83
3	38.8	28.55	20.75	18.05
4	41.55	30.5	20.38	21.17
5	43.45	31.8	20.37	23.08
6	44.0	32.2	20.20	23.80
7	45.5	33.0	20.07	25.43
8	46.0	33.6	19.97	26.03
9	46.55	34.05	19.82	26.73
10	46.9	35.5	20.10	26.80
11	47.95	36.65	21.00	26.95
12	49.0	37.9	22.85	26.15

Winding rise = $(R_2/R_1)(235+25.9) - 235 - 22.85 + 0.475$: HV 39.7 K, LV 34.3 K.

3.7 Pressure, vacuum and oil-leakage tests

Test	Condition	Result
Routine pressure	35 kPa, 10 min (14-11-2025)	No leakage at any point
Type pressure	80 kPa, 30 min	Max deflection 1.08 mm - Within limits
Vacuum	500 mmHg, 30 min	Max deflection 2.87 mm - Within limits
Oil leakage	28 kPa (2 x 14 kPa head), 8 h (13-11-2025)	No leakage at any point

4. Statement of conformity

Decision rule requested by the customer: Yes - (i) compliance per standard; MU reported if required. **The sample complied with all 23 automatically evaluated requirements (IS 1180 (latest: yes)).**

* Items flagged by automated validation - to be confirmed by the reviewer before approval:

Check	Detail
Identifier consistency	Losses logsheet (no-load): transcribed '25T1656' / 'HVD25S0847' but work instruction = 25T1654 / 0847. Verify handwriting (4/6/H).
Identifier consistency	Losses logsheet (resistance): transcribed '25T1656' / 'HVD25S0847' but work instruction = 25T1654 / 0847. Verify handwriting (4/6/H).
Identifier consistency	Routine test logsheet: transcribed '25T1654' / 'HVD25S0867' but work instruction = 25T1654 / 0847. Verify handwriting (4/6/H).
No-load watts sum	112.5%: $W_1+W_2+W_3 = 553.15$ but logged average/sum 553.73 (check reading)
HV winding temperature rise	39.7 K (limit 40 K, margin 0.3 K)
Reported vs computed oil rise	Logsheet reports 26.01 K, last-hour computed 26.15 K
Correction factor	Written as 0.275 at top of page 2 but 0.475 used in the formula - confirm

Test Engineer: P. Naveenkumar	Approved by: (pending)
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